NIALL CASSIDY

POST-PROTOTYPE

A2 WJEC COMPUTER SCIENCE

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Section 5: Post-Prototype (5/100) marks

•obtained feedback on the prototype system from a competent third party

• fully described the feedback received and explained the implications of the feedback for the re-design of the system • re-considered and re-structured input and output facilities as necessary

• given full consideration to the need for additional data as necessary

• given full consideration to all required files and/or data structures in light of any need to include additional data

• given full consideration to the need for any additional processes needed and has fully described any new relationships between the data and the processes of the revised design

• fully described all processing routines using an appropriate recognised convention in sufficient detail for implementation by a competent third party

• fully refined with detailed design, in response to feedback or justified instances where suggestions have been discounted.

5.0 Post-Prototype Discussion

To benefit from my time spent on the prototype even further, gaining feedback from competent third parties is invaluable to help my understanding and have direction going forward with the finished solution. To achieve this, I decided a simple online questionnaire would be sufficient, as it would be easy to share and fill out. I will use both qualitive and quantitative questions to gain both a general understanding of how the system is viewed and rated as well as the ability for more detailed feedback to be given. Answering these questions I have decided on three different perspectives that would give me a wide array of feedback. My classmate, the manager of Mulligan Motors, and a sales employee chosen at random.

The questions I have included are:

1) How would you rate the Prototype's overall responsiveness?

2) How would you rate the prototypes overall navigation features?

3) How would you rate the prototypes method of deleting/adding/editing a record?

4) How would you rate the prototypes login page?

5) Are there any features you feel are missing or could be added to the systems gui?

6) Do you think all data and fields in the database were appropriate to Mulligan Motors?

7) If no, do you have suggestions for fields to be deleted/ new fields to be added

8) Would you be happy with this new system replacing the old system?

9) Would you be happy with this new system replacing the old system?

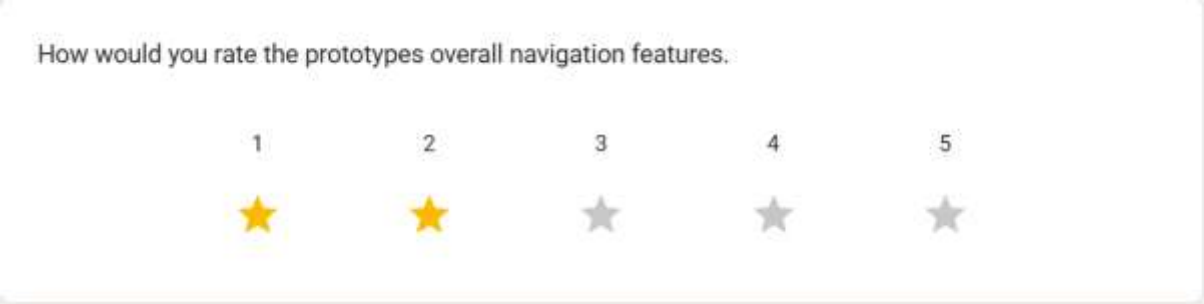
10) Do you feel this new system would require training to use fully?

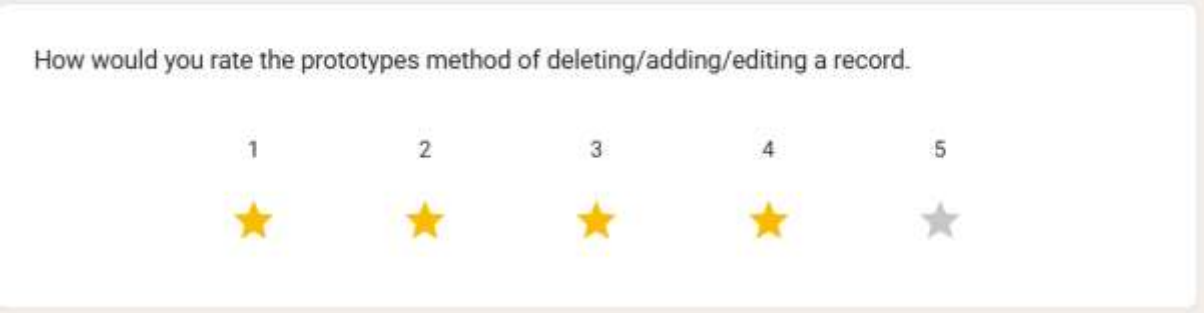
11) Is there anything else you would like to mention?

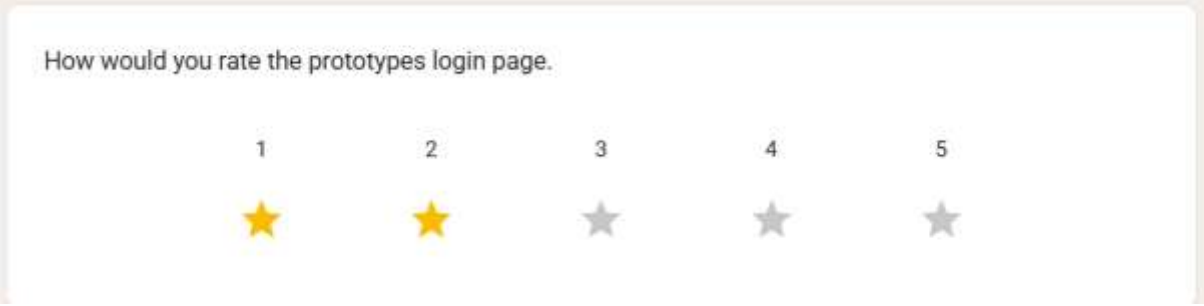
5.1 Overview and analysis of feedback

Feedback 1 : Staff member









Are there any features you feel are missing or could be added to the systems gui?

Login page window size needs to be expanded every time. This is because the login button cannot be seen. This is a big time waster.

Do you think all data and fields in the database were appropriate to Mulligan Motors?

Yes

Do you have suggestions for fields to be deleted/ new fields to be added?

Maybe date joined for employees. Allows me to easily give employment rewards.

Would you be happy with this new system replacing the old system?

- (●) Yes
- ( ) No

Do you feel this new system would require training to use fully?

- (●) Yes
- ( ) No

Is there anything else you would like to mention?

I feel as if the blue used in the GUI should be slightly brighter

Analysis of feedback 1:

Firstly, the staff member rated the system's responsiveness highly, indicating that the prototype operates efficiently without noticeable delays. This proves that the choice for SQlite3 to handle backend processes was a correct choice for this system and should be continued to be used. However, The login page received a low score due to a usability issue: the window size needs to be expanded every time to access the login button. This was described as a "big time waster," highlighting the need for immediate adjustment to the GUI design for this screen. This should be a simple fix in my annotated code as I can easily where screen size for login is determined.

The staff member also mentioned that the blue color used in the GUI could be "slightly brighter," indicating that the current color scheme feels dull. Enhancing the visual appeal with brighter tones or improved color contrast could make the interface more engaging and should be considered.

The staff member noted that the system would require training to use fully. This suggests that some features, especially navigation or record manipulation, might not be as intuitive as intended for users with varying levels of technical expertise. I can resolve this by adding a training manual to the system.

Finally, they recommended adding a "date joined" field for employees, which could help track employment duration and provide rewards more efficiently. This reflects a gap in the current data structure that could enhance the system's utility for human resource purposes.

Feedback 2: Manager

How would you rate the Prototype's overall responsiveness

1 2 3 4 5



How would you rate the prototypes overall navigation features.

1 2 3 4 5



How would you rate the prototypes method of deleting/adding/editing a record.

1 2 3 4 5



How would you rate the prototypes login page.

1 2 3 4 5



Are there any features you feel are missing or could be added to the systems gui?

Log off and previous window buttons.

Do you think all data and fields in the database were appropriate to Mulligan Motors?

Yes

Do you have suggestions for fields to be deleted/ new fields to be added?

Vehicle condition for tbl_VehicleStock.

Would you be happy with this new system replacing the old system?

- (●) Yes
- ( ) No

Do you feel this new system would require training to use fully?

- (●) Yes
- ( ) No

Is there anything else you would like to mention?

System not very secure - No encyrpytion or backups. Could add drop down boxes to certain fields

Analysis of feedback 2: Manager

The manager seems content in the prototype's ability to add/delete/edit records, navigate throughout the forms, and overall responsiveness as he rates them either 3 or 4 out of 5 stars. I am happy with this as these aspects will be marginally improved in the upcoming final solution. However, this reassuringly indicates no major restructuring of key system functions are required. The manager does suggest some improvements for navigation, asking for 'log off buttons and back buttons' I agree with his sentiment, as currently, you cannot log out of the prototype, requiring the program to be completely closed down and opened again for a new account. This is wholly inefficient and easy to implement so this is a very good point. I will also implement back buttons to go to a previous form. The prototype failed to implement this instead relying on operating systems built 'X' buttons (See example picture)



There is no way to leave the window other than pressing Windows 11 built in 'x' button.

For data, the manager suggested a "Vehicle condition for tbl_VehicleStock." I find this is a good idea as it allows the system to differentiate between extremely old cars and newer second-hand cars. This allows for overall more detail in the database system and is a field that should definitely be added to the final solution. Thankfully, he is happy with this system replacing the old traditional system from his answer but highlights it would need training to be used fully. In addition, he shows a concern for the lack of security built into the prototype. Whilst security was not an essential focus for this prototype system, I do understand his concerns and will try to incorporate a simple backup feature demonstrate one the of few simple security measures I plan to implement. Finally, drop down boxes were also proposed, I fully agree with this sentiment, and I think it would add greatly to the system. The presence of combo boxes would reduce the likelihood of typos therefore increasing data integrity.

Feedback 3: Classmate Dylan McDermot

How would you rate the Prototype's overall responsiveness

1 2 3 4 5



How would you rate the prototypes method of deleting/adding/editing a record.

1 2 3 4 5



How would you rate the prototypes login page.

1 2 3 4 5



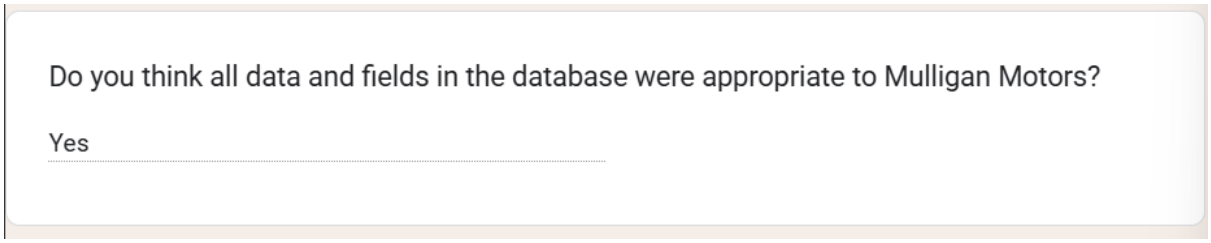
How would you rate the prototypes overall navigation features.

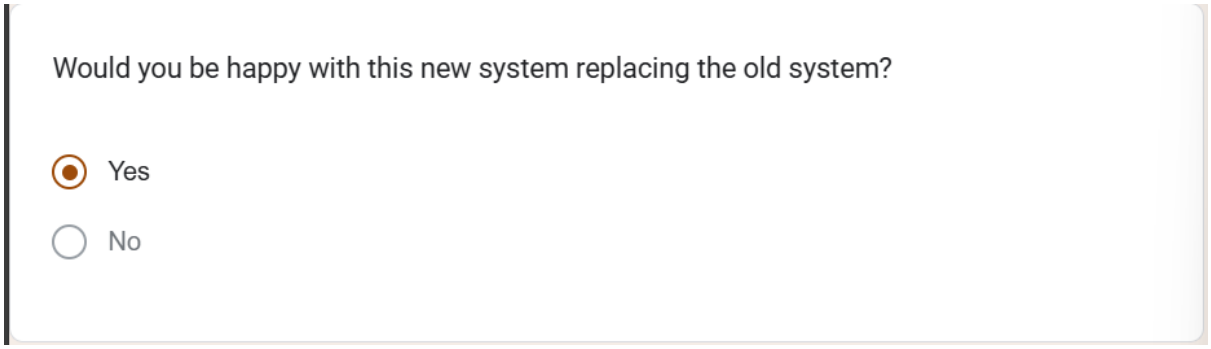
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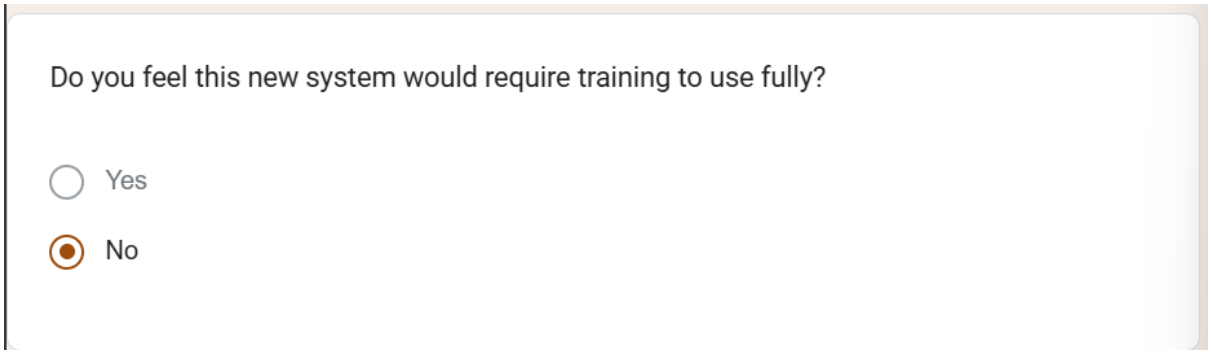


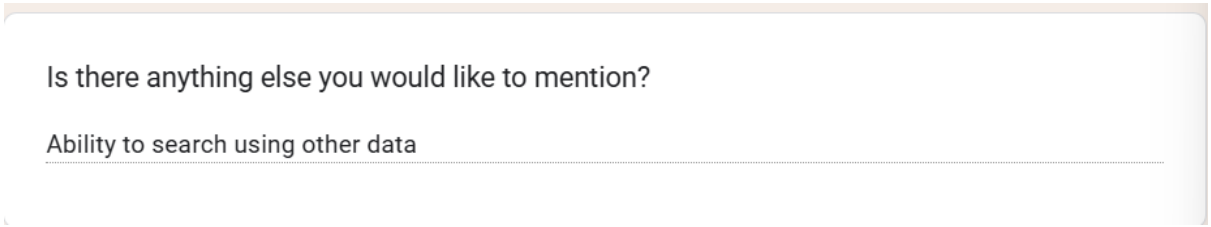
Are there any features you feel are missing or could be added to the systems gui?

Could add a help page









Analysis of Feedback 3

For core system functions, my classmate gave me ratings of 3-4 stars out of 5. I am happy with this for a prototype. He agreed that no data

items in the database need to be removed as they were all relevant. In addition, he responded that he would be happy for this system to replace Mulligan Motor's paper-based system.

However, my classmate still had suggestions to be implemented into the system. Even though he stated that the software would not require training for it to be utilised at its fullest potential, he recommended that a 'help page' be installed to help users quickly on the go within the prototype itself. Whilst I was hesitant to include this due to the already planned inclusion of a full training guide with the final solution and desire to keep the program small in space, I quickly realised that a user manual can potentially help returning users quickly come to grips with the system again. This would be especially useful if an employee went on an extended break for a holiday or maternity leave, for example. I will include this in the software. Finally, he mentions an issue with searching for items. He points out that searching for records only with an ID is impractical especially if the ID of a record is not known. He suggests that searches be made with other pieces of data. I concur with this idea and will add another layer of practicality to the software.

5.2 Updating the designs & system

Summary of changes to be made

1. Expand login window size.
2. Log Off button & Back Buttons
3. 'Date Joined' & 'Vehicle Condition' attributions
4. Search using data other than ID's
5. Help page
6. Brighter GUI colour
7. Drop down boxes
8. Simple backup

1. Expanding Login Window Size.

It was mentioned that the login screen had an issue with the automatic window size being too small for all features to show properly. As a result,

the button that needs to be pressed to actually process the username and password was cut off. This was confusing and unprofessional for the project, so I added size the login screen so that all features were included with plenty of room.

Here is the new code used in the login screen.



```
# Login screen
def login_screen():
    def validate_login():
        if username_var.get() == "admin" and password_var.get() == "1234":
            login_window.destroy()
            manager_home()
        else:
            messagebox.showerror("Login Failed", "Invalid username or password.")

    login_window = Tk()
    login_window.title("Mulligan Motors - Login")
    login_window.geometry("450x300")  # Increased by 50px in both width and height
    login_window.config(bg="#87CEEB")

    Label(login_window, text="Login to Mulligan Motors", font=("Arial", 16), bg="#87CEEB", fg="black").pack(pady=20)

    username_var = StringVar()
    password_var = StringVar()

    Label(login_window, text="Username", font=("Arial", 12), bg="#87CEEB", fg="black").pack(pady=5)
    Entry(login_window, textvariable=username_var, font=("Arial", 12)).pack(pady=5)

    Label(login_window, text="Password", font=("Arial", 12), bg="#87CEEB", fg="black").pack(pady=5)
    Entry(login_window, textvariable=password_var, show="*", font=("Arial", 12)).pack(pady=5)

    Button(login_window, text="Login", command=validate_login, font=("Arial", 12), bg="#4682B4", fg="white").pack(pady=20)

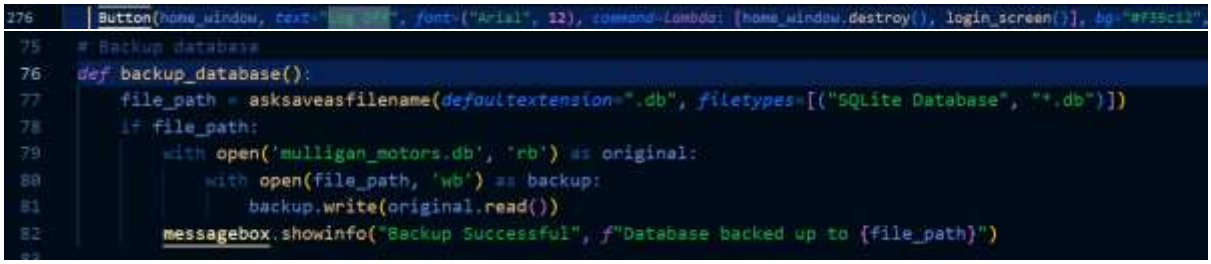
    login_window.mainloop()
```

Here is the result of running the new code.



2. Log off & Back buttons

It was identified that navigation and session management in the application were insufficiently intuitive, causing confusion for users. To address this, I implemented a "Log Off" button in the main menu and other forms to allow users to end their session securely. Additionally, "Back" buttons were added to individual forms, enabling users to navigate between screens seamlessly without needing to resort to pressing the built in window ‘x’s. These enhancements improved the application's usability and professionalism.





3. ‘Date Joined’ & ‘Vehicle Condition’ attributes

The system previously lacked key attributes that could enhance data relevance and functionality. To address this, the "Date Joined" attribute was added for customers to track when they registered with Mulligan Motors. Similarly, the "Vehicle Condition" attribute was introduced for vehicle records, enabling differentiation between new and used vehicles. These additions provide more granular data management and support better decision-making processes, such as marketing campaigns for long-term customers or inventory classification. This is a simple change and resultingly I feel evidence is not necessary.

4. Additional search options

The previous search functionality only allowed searches by ID, limiting its flexibility. A dropdown menu was added to let users select the field they want to search by, such as name, email, or ID. This addition not only improved the usability of the search function but also

aligned the system with real-world use cases, where searching by a variety of attributes is often necessary.

```
def search():
    search_field = dropdown_var.get()  # Get the selected field
    search_term = search_entry.get()  # Get the search term

    if not search_term:
        messagebox.showwarning("Invalid Input", "Please enter a search term.")
        return

    # Build query dynamically based on selected field
    query = f"SELECT * FROM {table_name} WHERE {search_field} LIKE ?"
    conn = sqlite3.connect('mulligan_motors.db')
    cursor = conn.cursor()
    cursor.execute(query, (f"%{search_term}%",))
    results = cursor.fetchall()
    conn.close()

    # Clear and populate Treeview
    for item in table_tree.get_children():
        table_tree.delete(item)

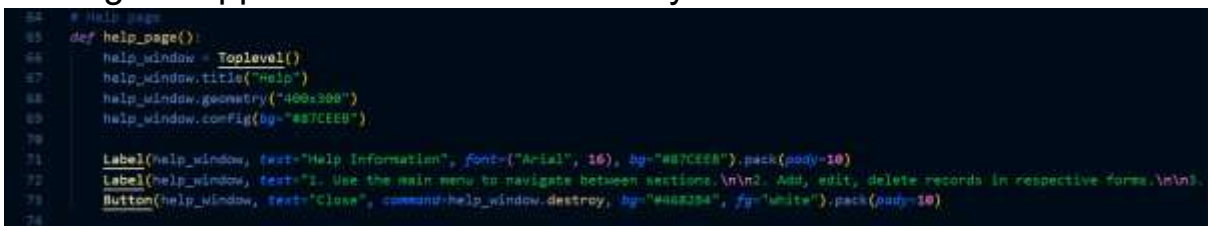
    if results:
        for result in results:
            table_tree.insert("", "end", values=result)
    else:
        messagebox.showinfo("No Records Found", f"No records found for {search_field} = '{search_term}'.")
```



5. Help page

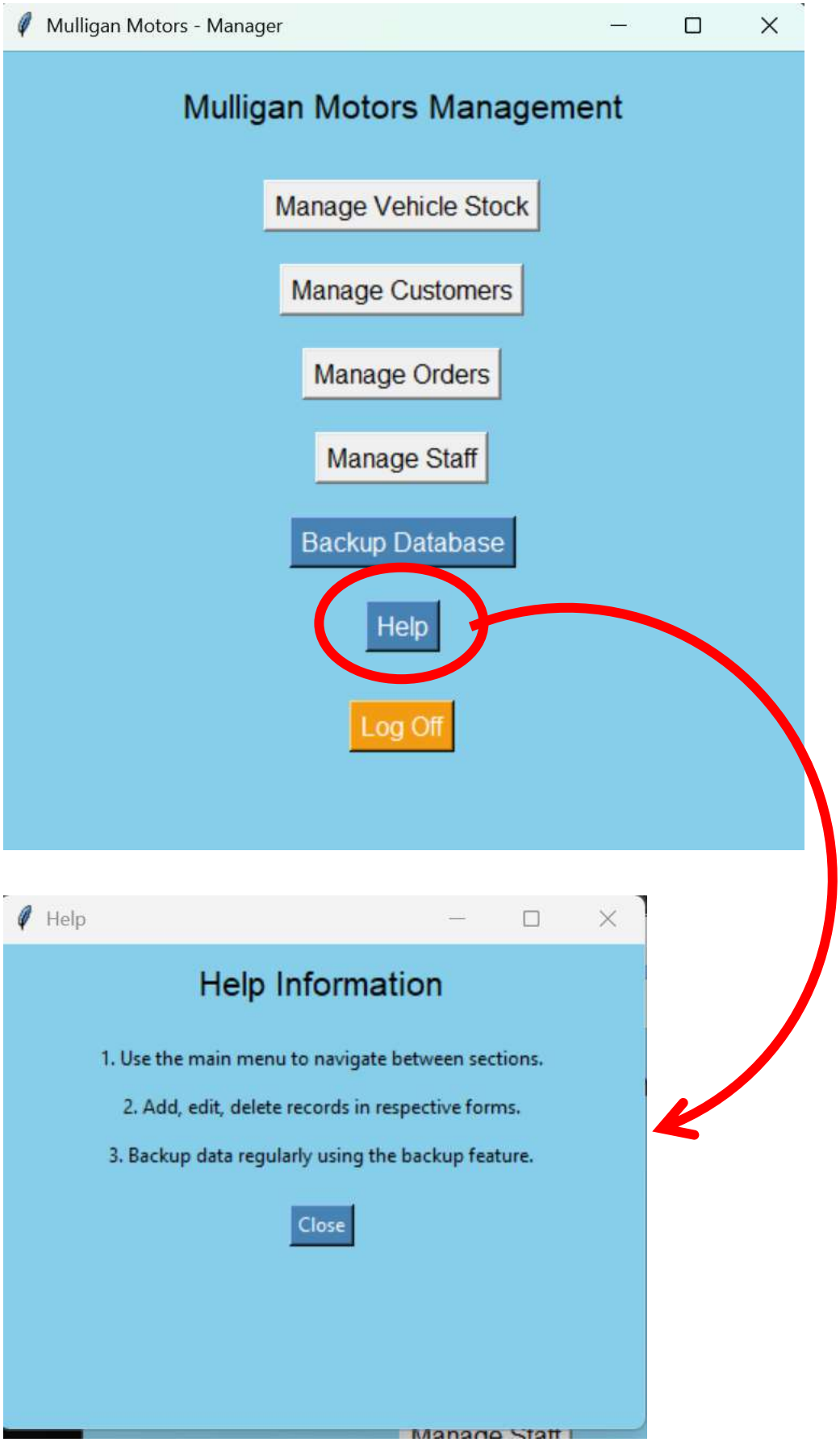
The absence of clear user guidance posed a challenge for navigating the application. A dedicated "Help" page was added, providing step-by-step instructions on how to use the main features, such as managing records, performing searches, and creating backups. This

feature enhanced user confidence and reduced the learning curve, making the application more user-friendly.



```
# Help page
def help_page():
    help_window = Toplevel()
    help_window.title("Help")
    help_window.geometry("400x300")
    help_window.config(bg="#87CEEB")

    Label(help_window, text="Help Information", font=("Arial", 16), bg="#87CEEB").pack(pady=10)
    Label(help_window, text="1. Use the main menu to navigate between sections.\n\n2. Add, edit, delete records in respective forms.\n\n3.
    Button(help_window, text="Close", command=help_window.destroy, bg="#4682B4", fg="white").pack(pady=10)
```


Mulligan Motors - Manager
Mulligan Motors Management
Manage Vehicle Stock
Manage Customers
Manage Orders
Manage Staff
Backup Database
Help
Log Off
Help
Help Information
1. Use the main menu to navigate between sections.
2. Add, edit, delete records in respective forms.
3. Backup data regularly using the backup feature.
Close

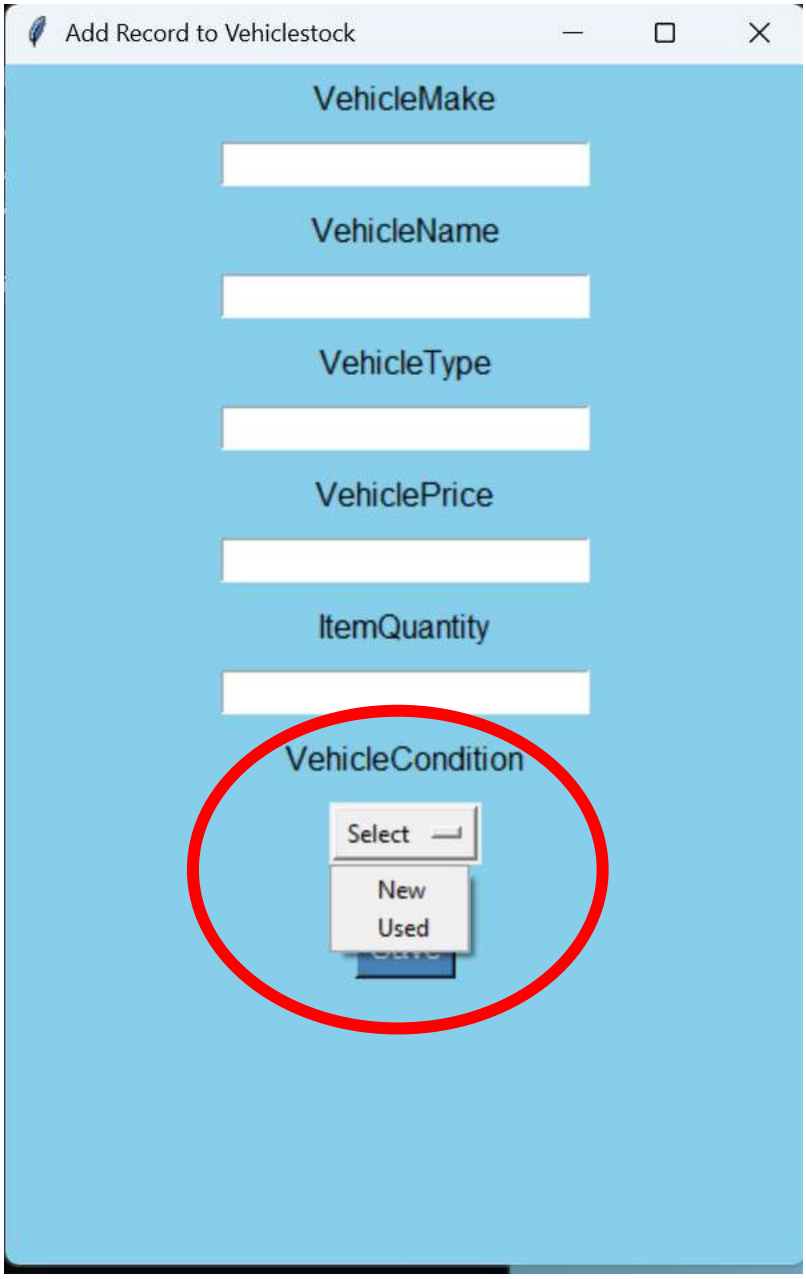
6. Brighter GUI Colour

The initial color scheme lacked visual appeal and clarity, which could negatively impact user experience. To improve aesthetics and readability, the GUI was redesigned with a brighter blue theme, featuring light blue backgrounds and darker blue buttons. This change not only improved the application's appearance but also enhanced its accessibility by creating better visual contrast. The hex code of the new background colour is *"#87CEEB"* and is visible in all the screenshots in this document.

7. Drop Down Boxes

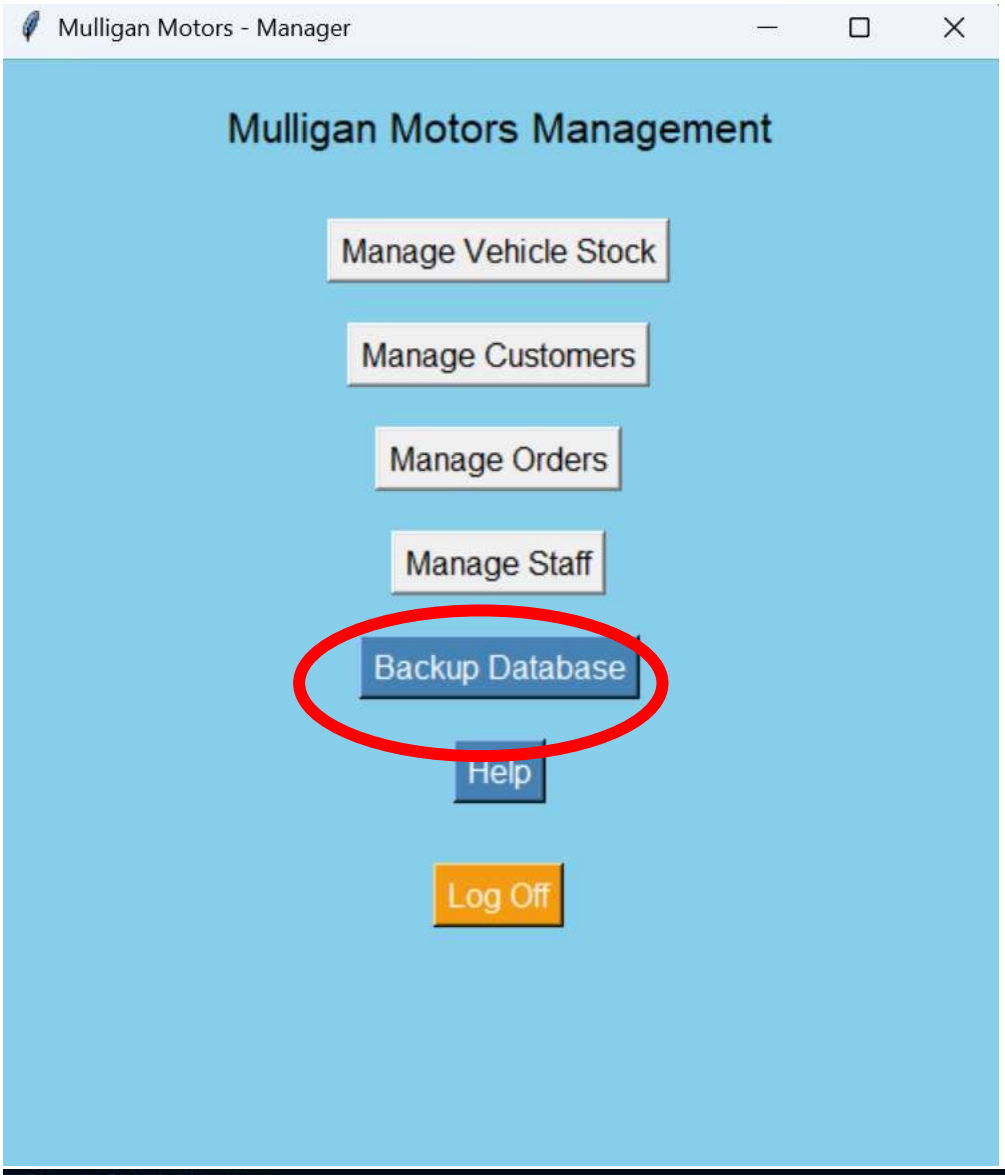
To simplify and standardize data entry, dropdown menus were integrated for specific input fields, such as "Vehicle Condition" and "Preferred Contact Method." Previously, users had to type these values manually, increasing the risk of input errors. The dropdown menus now restrict users to valid options, ensuring data consistency and minimizing mistakes during record creation or editing. I included two simple options for vehicle condition (New vs used) and three for contact preferences (Email, phone, or mail).



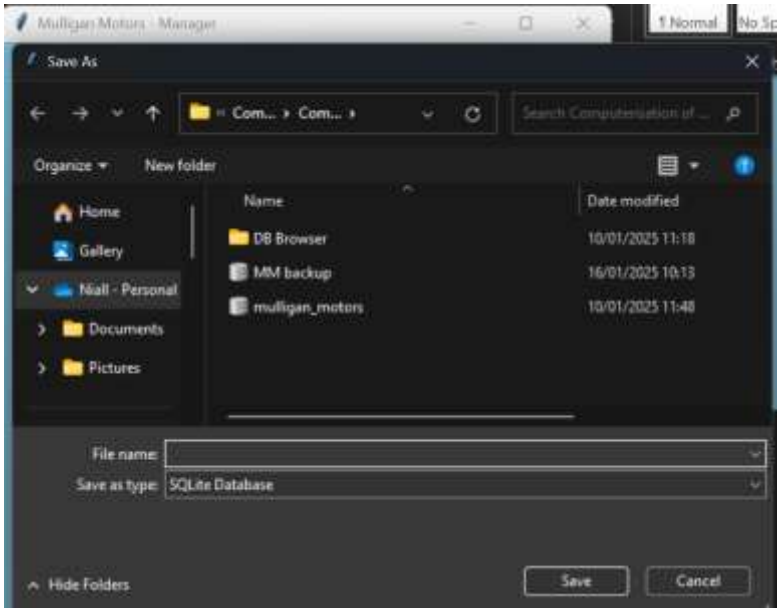


8. Simple backup feature.

It was noted that data security and reliability were potential risks without a proper backup mechanism by the system manager himself. To address this, a simple backup feature was implemented, allowing users to save the database to a file at any location of their choice. This feature ensures that data is protected against accidental loss, reinforcing the system's robustness.




```python
# Backup database
def backup_database():
    file_path = asksaveasfilename(defaultextension=".db", filetypes=[("SQLite Database", "*.db")])
    if file_path:
        with open('mulligan_motors.db', 'rb') as original:
            with open(file_path, 'wb') as backup:
                backup.write(original.read())
        messagebox.showinfo("Backup Successful", f"Database backed up to {file_path}")
```



A window will appear allowing the user to enter a file name and location which will save a backup copy. I named my backup 'MM Backup'

5.3 Success Criteria

After developing this prototype and carefully considering the feedback received, I remain confident in my ability to meet the original success criteria for the system. The progress made so far has demonstrated that the project is on track and achievable within the given timeframe. The prototype has proven to be functional and effective in core functions such as relationships, processes, and data storage. With such a strong foundation for the final program proven in the prototype, I am certain that I can deliver a finished project that excels in functionality and professionalism to Mulligan Motors. Overall, the program will fully align with the original objectives and user needs.